

Homework B

Name:

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1. Consider the function $f(x_1, x_2) = 5x_1^2 + x_2^2 + 4x_1x_2 - 14x_1 - 6x_2 + 20$.
 - (a) Write a py-file to use the steepest descent method to solve the above problem.
 - (b) Produce a table for the whole iterations with a random initial point.
 - (c) Write a py-file to plot the behavior of the steepest descent method by a contour map.
2. Consider the function

$$f(x) = x_1 - 0.6x_2 + 4x_3 + 0.25x_4 - \sum_{i=1}^4 \log(x_i) - \log\left(5 - \sum_{i=1}^4 x_i\right).$$

- (a) Write a py-file to use the steepest descent method to solve the above problem.
 - (b) Produce a table for the iterations: 1-10,20,30,40,50,100,110,120, with a random initial point.
3. Suppose that we seek to minimize

$$f(x) = \frac{1}{2}x^T Hx + c^T x,$$

where

$$H = \begin{pmatrix} 10 & -18 & 2 \\ -18 & 40 & -1 \\ 2 & -1 & 3 \end{pmatrix} \text{ and } c = \begin{pmatrix} 12 \\ -47 \\ -8 \end{pmatrix}.$$

Implement the steepest descent method on this problem, using the following starting points:

- $x_0 = (0, 0, 0)^T$.
- $x_0 = (15.09, 7.66, -6.56)^T$.
- $x_0 = (11.77, 6.42, -4.28)^T$.
- $x_0 = (4.46, 2.25, 1.85)^T$.

As it turns out, the optimal solution to this problem is $x^* = (4, 3, 1)^T$, with $f(x^*) = -50.5$.